

RADIOCARBON CALIBRATION PROGRAM*

CALIB REV8.2

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*To be used in conjunction with:

Stuiver, M., and Reimer, P.J., 1993, Radiocarbon, 35, 215-230.

KKPP24_1_1

73631

sediment

Radiocarbon Age BP 1120 +/- 20

Calibration data set: intcal20.14c

% area enclosed cal BP age ranges

68.3 (1 sigma) cal BP 974 - 1008
1020 - 1038
1047 - 1057

95.4 (2 sigma) cal BP 960 - 1061

Median Probability: 1012

Reimer et al. 2020
relative area under
probability distribution

0.559

0.268

0.172

1.000

KKPP24_1_5

73632

sediment

Radiocarbon Age BP 4340 +/- 25

Calibration data set: intcal20.14c

% area enclosed cal BP age ranges

68.3 (1 sigma) cal BP 4853 - 4884
4900 - 4909
4930 - 4959

95.4 (2 sigma) cal BP 4848 - 4966

Median Probability: 4903

Reimer et al. 2020
relative area under
probability distribution

0.528

0.076

0.396

1.000

KKPP24_1_1

73633

sediment

Radiocarbon Age BP 3090 +/- 20

Calibration data set: intcal20.14c

% area enclosed cal BP age ranges

68.3 (1 sigma) cal BP 3255 - 3293
3329 - 3357

95.4 (2 sigma) cal BP 3239 - 3367

Median Probability: 3294

Reimer et al. 2020
relative area under
probability distribution

0.591

0.409

1.000

KKPP24_2_1

73634

sediment

Radiocarbon Age BP 1530 +/- 20

Calibration data set: intcal20.14c

% area enclosed cal BP age ranges

68.3 (1 sigma) cal BP 1373 - 1408

95.4 (2 sigma) cal BP 1351 - 1416

1454 - 1472

1490 - 1495

1499 - 1510

Median Probability: 1394

Reimer et al. 2020
relative area under
probability distribution

1.000

0.900

0.063

0.009

0.028

KKPP24_2_4

73635

sediment

Radiocarbon Age BP 4130 +/- 20

Calibration data set: intcal20.14c

% area enclosed cal BP age ranges

68.3 (1 sigma)	cal BP 4580 - 4601
	4612 - 4648
	4674 - 4697
	4757 - 4802
95.4 (2 sigma)	cal BP 4532 - 4547
	4571 - 4658
	4662 - 4725
	4751 - 4817

Median Probability: 4677

Reimer et al. 2020

relative area under
probability distribution

0.161
0.296
0.192
0.351
0.024
0.426
0.234
0.315

KKPP24_2_8

73636

sediment

Radiocarbon Age BP 7880 +/- 30

Calibration data set: intcal20.14c

% area enclosed cal BP age ranges

68.3 (1 sigma)	cal BP 8598 - 8652
	8668 - 8719
95.4 (2 sigma)	cal BP 8590 - 8778
	8833 - 8860
	8889 - 8890
	8919 - 8937
	8960 - 8972

Median Probability: 8675

Reimer et al. 2020

relative area under
probability distribution

0.574
0.426
0.943
0.033
0.001
0.015
0.008

KKPP24_3_1

73637

sediment

Radiocarbon Age BP 950 +/- 20

Calibration data set: intcal20.14c

% area enclosed cal BP age ranges

68.3 (1 sigma)	cal BP 798 - 813
	816 - 837
	842 - 869
	901 - 910
95.4 (2 sigma)	cal BP 793 - 890
	897 - 917

Median Probability: 848

Reimer et al. 2020

relative area under
probability distribution

0.199
0.287
0.386
0.128
0.831
0.169

KKPP24_3_5

73638

sediment

Radiocarbon Age BP 2000 +/- 20

Calibration data set: intcal20.14c

% area enclosed cal BP age ranges

68.3 (1 sigma)	cal BP 1893 - 1903
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Reimer et al. 2020

relative area under
probability distribution

0.139

	1923 - 1947	0.420
	1961 - 1987	0.441
95.4 (2 sigma)	cal BP 1880 - 1994	1.000
Median Probability: 1940		
KKPP24_3_9		
73639		
sediment		
Radiocarbon Age BP 3850 +/- 20		
Calibration data set: intcal20.14c		
% area enclosed	cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP 4159 - 4168	0.067
	4182 - 4199	0.146
	4232 - 4294	0.675
	4333 - 4348	0.113
95.4 (2 sigma)	cal BP 4154 - 4206	0.237
	4222 - 4305	0.534
	4308 - 4312	0.005
	4322 - 4357	0.137
	4367 - 4404	0.087
Median Probability: 4265		
KKPP24_4_1		
73640		
sediment		
Radiocarbon Age BP 490 +/- 20		
Calibration data set: intcal20.14c		
% area enclosed	cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP 512 - 528	1.000
95.4 (2 sigma)	cal BP 506 - 539	1.000
Median Probability: 521		
KKPP24_4_5		
73641		
sediment		
Radiocarbon Age BP 2820 +/- 30		
Calibration data set: intcal20.14c		
% area enclosed	cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP 2878 - 2956	1.000
95.4 (2 sigma)	cal BP 2848 - 3004	0.989
	3019 - 3027	0.006
	3047 - 3055	0.005
Median Probability: 2921		
KKPP24_4_9		
73642		
sediment		
Radiocarbon Age BP 8290 +/- 20		
Calibration data set: intcal20.14c		
% area enclosed	cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP 9152 - 9167	0.105
	9272 - 9319	0.438
	9357 - 9404	0.457

95.4 (2 sigma)	cal BP 9142 - 9177	0.131
	9202 - 9238	0.057
	9244 - 9333	0.410
	9341 - 9418	0.401
Median Probability: 9307		
KKPP24_5_4		
73644		
sediment		
Radiocarbon Age BP 3050 +/- 20		
Calibration data set: intcal20.14c		
% area enclosed	cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP 3217 - 3257	0.511
	3291 - 3330	0.489
95.4 (2 sigma)	cal BP 3177 - 3195	0.045
	3206 - 3276	0.502
	3281 - 3346	0.453
Median Probability: 3264		
KKPP24_5_6		
73645		
sediment		
Radiocarbon Age BP 4110 +/- 25		
Calibration data set: intcal20.14c		
% area enclosed	cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP 4531 - 4552	0.135
	4570 - 4624	0.455
	4637 - 4643	0.031
	4679 - 4692	0.083
	4761 - 4796	0.297
95.4 (2 sigma)	cal BP 4524 - 4654	0.606
	4667 - 4707	0.127
	4753 - 4812	0.267
Median Probability: 4628		
KKPP24_6_1		
73646		
sediment		
Radiocarbon Age BP 950 +/- 20		
Calibration data set: intcal20.14c		
% area enclosed	cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP 798 - 813	0.199
	816 - 837	0.287
	842 - 869	0.386
	901 - 910	0.128
95.4 (2 sigma)	cal BP 793 - 890	0.831
	897 - 917	0.169
Median Probability: 848		
KKPP24_6_6		
73647		
sediment		
Radiocarbon Age BP 2370 +/- 20		
Calibration data set: intcal20.14c		
# Reimer et al. 2020		

% area enclosed	cal BP age ranges	relative area under probability distribution
68.3 (1 sigma)	cal BP 2345 - 2367	0.748
	2390 - 2408	0.252
95.4 (2 sigma)	cal BP 2342 - 2437	0.958
	2448 - 2464	0.042
Median Probability: 2365		

KKPP24_6_1
73648
sediment
Radiocarbon Age BP 6510 +/- 35
Calibration data set: intcal20.14c

% area enclosed	cal BP age ranges	relative area under probability distribution
68.3 (1 sigma)	cal BP 7336 - 7382	0.565
	7420 - 7431	0.232
	7449 - 7467	0.202
95.4 (2 sigma)	cal BP 7325 - 7401	0.512
	7412 - 7500	0.488
Median Probability: 7405		

KKPP24_7_1
73649
sediment
Radiocarbon Age BP 550 +/- 20
Calibration data set: intcal20.14c

% area enclosed	cal BP age ranges	relative area under probability distribution
68.3 (1 sigma)	cal BP 529 - 554	0.935
	617 - 621	0.065
95.4 (2 sigma)	cal BP 526 - 556	0.717
	597 - 625	0.283
Median Probability: 548		

KKPP24_7_5
73650
sediment
Radiocarbon Age BP 3250 +/- 20
Calibration data set: intcal20.14c

% area enclosed	cal BP age ranges	relative area under probability distribution
68.3 (1 sigma)	cal BP 3416 - 3418	0.043
	3447 - 3481	0.957
95.4 (2 sigma)	cal BP 3396 - 3434	0.228
	3438 - 3491	0.758
	3541 - 3548	0.013
Median Probability: 3459		

KKPP24_7_8
73651
sediment
Radiocarbon Age BP 3420 +/- 20
Calibration data set: intcal20.14c

% area enclosed	cal BP age ranges	relative area under probability distribution
68.3 (1 sigma)	cal BP 3594 - 3600	0.066

		3634 - 3696	0.934
95.4 (2 sigma)	cal BP	3579 - 3609	0.124
		3614 - 3718	0.826
		3800 - 3816	0.050
Median Probability: 3662			
KKPP24_8_1			
73652			
sediment			
Radiocarbon Age BP 960 +/- 20			
Calibration data set: intcal20.14c			
% area enclosed		cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP	800 - 811	0.147
		823 - 850	0.419
		853 - 868	0.224
		902 - 916	0.211
95.4 (2 sigma)	cal BP	793 - 875	0.772
		883 - 889	0.014
		897 - 922	0.214
Median Probability: 848			
KKPP24_8_4			
73653			
sediment			
Radiocarbon Age BP 2020 +/- 20			
Calibration data set: intcal20.14c			
% area enclosed		cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP	1938 - 1952	0.280
		1956 - 1992	0.720
95.4 (2 sigma)	cal BP	1889 - 1913	0.071
		1919 - 2000	0.929
Median Probability: 1961			
KKPP24_8_7			
73654			
sediment			
Radiocarbon Age BP 8600 +/- 20			
Calibration data set: intcal20.14c			
% area enclosed		cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP	9537 - 9547	1.000
95.4 (2 sigma)	cal BP	9530 - 9557	0.945
		9574 - 9597	0.055
Median Probability: 9544			
KKPP24_4_7			
73655			
charcoal			
Radiocarbon Age BP 1180 +/- 20			
Calibration data set: intcal20.14c			
% area enclosed		cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP	1063 - 1081	0.298
		1091 - 1122	0.488
		1163 - 1175	0.214

95.4 (2 sigma)	cal BP 1006 - 1022	0.041
	1057 - 1152	0.781
	1159 - 1176	0.178
Median Probability: 1104		
KKPP24_7_7		
73656		
charcoal		
Radiocarbon Age BP 1140 +/- 20		
Calibration data set: intcal20.14c		
% area enclosed	cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP 977 - 983	0.078
	992 - 1032	0.702
	1051 - 1065	0.221
95.4 (2 sigma)	cal BP 961 - 967	0.012
	971 - 1073	0.927
	1105 - 1117	0.023
	1166 - 1175	0.037
Median Probability: 1021		
KKPP24_8_3		
73657		
charcoal		
Radiocarbon Age BP 270 +/- 20		
Calibration data set: intcal20.14c		
% area enclosed	cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP 292 - 315	0.829
	412 - 420	0.171
95.4 (2 sigma)	cal BP 156 - 165	0.035
	286 - 320	0.645
	378 - 386	0.017
	390 - 426	0.303
Median Probability: 309		
KKPP24_7_6		
73658		
charcoal		
Radiocarbon Age BP 1230 +/- 20		
Calibration data set: intcal20.14c		
% area enclosed	cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP 1080 - 1091	0.112
	1122 - 1163	0.681
	1175 - 1177	0.024
	1227 - 1243	0.183
95.4 (2 sigma)	cal BP 1070 - 1167	0.760
	1174 - 1177	0.017
	1210 - 1246	0.219
	1256 - 1258	0.004
Median Probability: 1146		
KKPP24_7_4		
73659		
charcoal		
Radiocarbon Age BP 1210 +/- 20		

Calibration data set: intcal20.14c
 % area enclosed cal BP age ranges

68.3 (1 sigma)	cal BP 1075 - 1100
	1119 - 1155
	1157 - 1164
95.4 (2 sigma)	cal BP 1065 - 1177
	1238 - 1241

Median Probability: 1126

Reimer et al. 2020
 relative area under
 probability distribution

0.373
0.521
0.106
0.994
0.006

KKPP24_6_8
 73660
 charcoal
 Radiocarbon Age BP 1270 +/- 25
 Calibration data set: intcal20.14c
 % area enclosed cal BP age ranges

68.3 (1 sigma)	cal BP 1176 - 1178
	1184 - 1190
	1206 - 1267
95.4 (2 sigma)	cal BP 1129 - 1140
	1145 - 1160
	1175 - 1282

Median Probability: 1225

Reimer et al. 2020
 relative area under
 probability distribution

0.022
0.081
0.897
0.027
0.039
0.934

KKPP24_7_7
 73661
 charcoal
 Radiocarbon Age BP 1210 +/- 20
 Calibration data set: intcal20.14c
 % area enclosed cal BP age ranges

68.3 (1 sigma)	cal BP 1075 - 1100
	1119 - 1155
	1157 - 1164
95.4 (2 sigma)	cal BP 1065 - 1177
	1238 - 1241

Median Probability: 1126

Reimer et al. 2020
 relative area under
 probability distribution

0.373
0.521
0.106
0.994
0.006

KKPP24_3_1
 73662
 charcoal
 Radiocarbon Age BP 860 +/- 20
 Calibration data set: intcal20.14c
 % area enclosed cal BP age ranges

68.3 (1 sigma)	cal BP 731 - 750
	756 - 774
95.4 (2 sigma)	cal BP 698 - 701
	723 - 792

Median Probability: 757

Reimer et al. 2020
 relative area under
 probability distribution

0.529
0.471
0.003
0.997

KKPP24_5_5
 73663
 charcoal
 Radiocarbon Age BP 1240 +/- 20
 Calibration data set: intcal20.14c

Reimer et al. 2020

% area enclosed	cal BP age ranges	relative area under probability distribution
68.3 (1 sigma)	cal BP 1127 - 1161	0.501
	1175 - 1177	0.009
	1212 - 1245	0.490
95.4 (2 sigma)	cal BP 1073 - 1104	0.102
	1113 - 1164	0.422
	1174 - 1178	0.024
	1185 - 1189	0.008
	1207 - 1267	0.445
Median Probability: 1161		

KKPP24_3_6
73664
charcoal
Radiocarbon Age BP 870 +/- 20
Calibration data set: intcal20.14c

% area enclosed	cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP 734 - 752	0.450
	754 - 775	0.550
95.4 (2 sigma)	cal BP 727 - 793	0.987
	891 - 897	0.013
Median Probability: 761		

KKPP24_3_6
73665
charcoal
Radiocarbon Age BP 940 +/- 20
Calibration data set: intcal20.14c

% area enclosed	cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP 794 - 835	0.524
	843 - 859	0.202
	862 - 873	0.137
	898 - 909	0.137
95.4 (2 sigma)	cal BP 792 - 913	1.000
Median Probability: 850		

KKPP24_6_1
73667
charcoal
Radiocarbon Age BP 100 +/- 20
Calibration data set: intcal20.14c

% area enclosed	cal BP age ranges	# Reimer et al. 2020 relative area under probability distribution
68.3 (1 sigma)	cal BP 38 - 70	0.371
	114 - 137	0.291
	227 - 253	0.338
95.4 (2 sigma)	cal BP 32 - 140	0.733
	224 - 257	0.267
Median Probability: 115		

KKPP24_5_3
73668
charcoal
Radiocarbon Age BP 1130 +/- 20

Calibration data set: intcal20.14c		# Reimer et al. 2020
% area enclosed	cal BP age ranges	relative area under probability distribution
68.3 (1 sigma)	cal BP 976 - 985	0.152
	990 - 1032	0.699
	1051 - 1060	0.149
95.4 (2 sigma)	cal BP 960 - 968	0.027
	970 - 1066	0.968
	1171 - 1173	0.005
Median Probability: 1017		

References for calibration datasets:

Reimer P, Austin WEN, Bard E, Bayliss A, Blackwell PG, Bronk Ramsey C, Butzin M, Edwards RL, Friedrich M, Grootes PM, Guilderson TP, Hajdas I, Heaton TJ, Hogg A, Kromer B, Manning SW, Muscheler R, Palmer JG, Pearson C, van der Plicht J, Reim, Richards DA, Scott EM, Southon JR, Turney CSM, Wacker L, Adolphi F, Büntgen U, Fahrni S, Fogtmann-Schulz A, Friedrich R, Köhler P, Kudsk S, Miyake F, Olsen J, Sakamoto M, Sookdeo A, Talamo S. 2020. The IntCal20 Northern Hemisphere radiocarbon age calibration curve (0-55 cal kB Radiocarbon 62. doi: 10.1017/RDC.2020.41.

Comments:

* This standard deviation (error) includes a lab error multiplier.
 ** 1 sigma = square root of (sample std. dev.^2 + curve std. dev.^2)
 ** 2 sigma = 2 x square root of (sample std. dev.^2 + curve std. dev.^2)
 where ^2 = quantity squared.
 [] = calibrated range impinges on end of calibration data set
 0* represents a "negative" age BP
 1955* or 1960* denote influence of nuclear testing C-14

NOTE: Cal ages and ranges are rounded to the nearest year which may be too precise in many instances. Users are advised to round results to the nearest 10 yr for samples with standard deviation in the radiocarbon age greater than 50 yr.